

Basic Skill	α (std err)	β (std err)	ζ (std err)
<i>All skill importance scores are normalized to be between 0 and 1.</i>			
Constant	0.082*** (0.011)	-0.112*** (0.011)	0.300*** (0.057)
Active Listening	0.128** (0.047)	0.214*** (0.043)	0.449*** (0.027)
Mathematics	-0.127*** (0.026)	0.161*** (0.021)	0.787*** (0.049)
Reading Comprehension	0.153*** (0.041)	0.470*** (0.037)	-0.346*** (0.017)
Science	-0.114*** (0.014)	-0.230*** (0.012)	-0.346*** (0.017)
Speaking	-0.028 (0.039)	0.133*** (0.033)	0.294*** (0.042)
Writing	0.368*** (0.042)	0.467*** (0.037)	0.566*** (0.047)
Active Learning	-0.157*** (0.027)	-0.065** (0.024)	0.028 (0.032)
Critical Thinking	-0.264*** (0.036)	-0.196*** (0.033)	-0.129** (0.042)
Learning Strategies	-0.072* (0.028)	-0.209*** (0.025)	-0.346*** (0.034)
Monitoring	-0.067** (0.023)	-0.149*** (0.020)	-0.232*** (0.026)
Programming	0.637*** (0.030)	0.623*** (0.022)	0.609*** (0.024)

Table 5: Regression of occupation-level, human-annotated exposure to GPTs on skill importance for each skill in the O*NET Basic skills category, plus the programming skill. Descriptions of the skills may be found in Appendix B. Task ratings within each occupation for exposure have equal weight.

Job Zone	Preparation Required	Education Required	Example Occupations	Median Income	Tot Emp (000s)	H α	M α	H β	M β	H ζ	M ζ
1	None or little (0-3 months)	High school diploma or GED (optional)	Food preparation workers, dishwashers, floor sanders	\$30,230	13,100	0.03	0.04	0.06	0.06	0.09	0.08
2	Some (3-12 months)	High school diploma	Orderlies, customer service representatives, tellers	\$38,215	73,962	0.07	0.12	0.16	0.20	0.24	0.27
3	Medium (1-2 years)	Vocational school, on-the-job training, or associate's degree	Electricians, barbers, medical assistants	\$54,815	37,881	0.11	0.14	0.26	0.32	0.41	0.51
4	Considerable (2-4 years)	Bachelor's degree	Database administrators, graphic designers, cost estimators	\$77,345	56,833	0.23	0.18	0.47	0.51	0.71	0.85
5	Extensive (4+ years)	Master's degree or higher	Pharmacists, lawyers, astronomers	\$81,980	21,221	0.23	0.13	0.43	0.45	0.63	0.76

Table 6: Mean exposure to GPTs by job zone. For each job zone, we also present the median of median annual income for each constituting occupation in USD, and the total number of workers in all occupations for that job zone, in the thousands. Task weights are equal for all tasks.

On The Job Training Required	Median Income	Tot Emp (000s)	H α	M α	H β	M β	H ζ	M ζ
None	\$77,440	90,776	0.20	0.16	0.42	0.46	0.63	0.76
Apprenticeship	\$55,995	3,066	0.01	0.02	0.04	0.06	0.07	0.10
Internship/residency	\$77,110	3,063	0.16	0.06	0.36	0.38	0.55	0.71
Short-term on-the-job training	\$33,370	66,234	0.11	0.15	0.21	0.25	0.32	0.34
Moderate-term on-the-job training	\$46,880	31,285	0.09	0.12	0.21	0.25	0.32	0.38
Long-term on-the-job training	\$48,925	5,070	0.08	0.10	0.18	0.22	0.28	0.33

Table 7: Mean exposure scores for occupations, grouped by level of on-the-job training required to attain competency in the job. Alongside exposure scores, we display the median of median annual income for each occupation, as well as the total number of workers in each group, in thousands. Task weights are equal within an occupation and sum to one.

5 Validation of Measures

5.1 Comparison to Earlier Efforts

This paper aims to build on a number of previous empirical studies examining the occupational exposure to advances in AI and/or automation. Previous studies have used a variety of methods, including:

- Using occupational taxonomies like O*NET to characterize which occupations have routine vs. non-routine and manual vs. cognitive task content (Autor et al., 2003; Acemoglu and Autor, 2011a).
- Mapping text descriptions of tasks to descriptions of technological advances in patents. (Kogan et al., 2021; Webb, 2020)
- Linking capabilities of AI systems to occupational abilities and aggregating exposure estimates to the occupations where those abilities are required. (Felten et al., 2018, 2023)
- Mapping the results of AI task benchmark evaluations (ImageNet, Robocup, etc.) to 59 worker tasks through a set of 14 cognitive abilities drawn from the cognitive science literature. (Tolan et al., 2021)
- Expert labeling of automation potential for a set of O*NET occupations where experts had high confidence, combined with a probabilistic classifier to estimate automation potential for the remainder of O*NET occupations. (Frey and Osborne, 2017)
- Developing a rubric for evaluating the "suitability for machine learning" (SML) of activities that workers are completing in the economy (Brynjolfsson and Mitchell, 2017; Brynjolfsson et al., 2018, 2023).

We provide a set of summary statistics on many of these prior efforts in Table 8.

This paper's methodology primarily builds upon the SML approach by developing a rubric to evaluate the overlap between LLM capabilities and worker tasks as reported in the O*NET database. Table 9 presents the results of OLS regressions of our new LLM exposure measurements on occupation-level exposure measures from (Felten et al., 2018) ("AI Occupational Exposure Score" in the table), (Frey and Osborne, 2017) (Frey & Osborne Automation), scores from all three technologies in (Webb, 2020), normalized routine manual and cognitive scores from (Acemoglu and Autor, 2011a), and (Brynjolfsson et al., 2018, 2023) (SML). We also use annualized occupational salaries from the most recent BLS Occupational Employment Survey as a control. There are four separate output variables representing new scores in this paper that are predicted by earlier efforts.

GPT-4 Exposure Rating 1 corresponds to our overall exposure rubric as evaluated by GPT-4, where full exposure potential is coded as 1, no exposure potential is coded as 0, and partial exposure (E2 in our labeling scheme) is coded as 0.5. GPT-4 Exposure Rating 2 is scored similarly for overall exposure, but with a slightly different prompt. The results are very similar across the two prompts. Human Exposure Rating represents the same rubric as in GPT-4 Exposure Rating 1 but is scored by humans, as discussed in an earlier section of the paper. These results correspond to the β set of statistics presented above, with supplemental tasks having half the weight of core tasks within an occupation. These weights sum to one (core/supplemental distinctions are determined by O*NET).

The results across each type of measurement are consistent. We find generally positive and statistically significant correlations between our LLM exposure measures and previous measurements targeting software and AI. Encouragingly, the SML exposure scores by occupation show significant and positive associations with the exposure scores we develop in this paper, demonstrating a level of cohesion between the two studies with similar approaches. The Webb software and AI patent-based measures, SML, and normalized (demeaned

	Min	25th Perc.	Median	75th Perc	Max	Mean	Std. Dev.	Count
GPT-4 Exposure Rating 1	0.00	0.13	0.34	0.50	1.00	0.33	0.22	750
GPT-4 Exposure Rating 2	0.00	0.09	0.24	0.40	0.98	0.26	0.20	750
Human Exposure Rating	0.00	0.09	0.29	0.47	0.84	0.29	0.21	750
Software (Webb)	1.00	25.00	50.00	75.00	100.00	50.69	30.05	750
Robot (Webb)	1.00	22.00	52.00	69.00	100.00	48.61	28.61	750
AI (Webb)	1.00	28.00	55.00	82.00	100.00	54.53	29.65	750
Suitability for Machine Learning	2.60	2.84	2.95	3.12	3.55	2.99	0.18	750
Normalized Routine Cognitive	-3.05	-0.46	0.10	0.63	3.42	0.07	0.86	750
Normalized Routine Manual	-1.81	-0.81	-0.11	0.73	2.96	0.05	1.01	750
AI Occupational Exposure Score	1.42	3.09	3.56	4.04	6.54	3.56	0.70	750
Frey & Osborne Automation	0.00	0.07	0.59	0.88	0.99	0.50	0.38	681
Log Avg. Salary	10.13	10.67	11.00	11.34	12.65	11.02	0.45	749

Table 8: Summary statistics for a suite of prior efforts to measure occupational exposure to AI and automation. We have also included summary statistics for measurements newly presented in this work. We include all measures from (Webb, 2020), normalized routine cognitive and manual scores from (Acemoglu and Autor, 2011a) (means may deviate slightly from 0 due to imperfect matching of occupational groups), Suitability for Machine Learning from (Brynjolfsson and Mitchell, 2017; Brynjolfsson et al., 2018, 2023), AI Occupational Exposure from (Felten et al., 2018), and Automation exposure from (Frey and Osborne, 2017). We include as many occupations as we can match, but since O*NET taxonomies have changed as these measures have been developed, some of the roles may be missing from the most recent version of O*NET 6-digit occupations.

and divided by standard deviation) routine cognitive scores all exhibit positive associations with some of our measures.

Software, SML, and routine cognitive scores all show positive and statistically significant associations with LLM exposure scores at a 1% level. Coefficients on AI scores from (Webb, 2020) are also positive and statistically significant at a 5% level, but our secondary prompt on overall exposure to LLMs in columns 3 and 4 does not exhibit a statistically significant relationship. For the most part, the AI Occupational Exposure Score is not correlated with our exposure measures. Webb’s Robot exposure scores, routine manual task content, and the overall Automation metric from (Frey and Osborne, 2017) are all negatively correlated with our primary GPT-4 and human-assessed overall exposure ratings, conditional on the other measurements. This negative correlation reflects the limited exposure of physical tasks to LLMs. Manual work is not exposed to LLMs or even LLMs with additional systems integration for the time being.

Low correlations with (Felten et al., 2018) and (Frey and Osborne, 2017) could potentially be explained by differences in approaches. Linking AI capabilities to worker abilities or scoring exposure directly based on the occupation’s characteristics, rather than aggregating up to the occupation from DWA or task-level scoring (as in the SML paper and our own), offer a slightly different perspective on the content of occupations.

In all regressions, the R^2 ranges between 60.7% (column 3) and 72.8% (column 5). This suggests that our measure, which explicitly focuses on LLM capabilities, has between 28 and 40% unexplained variance compared to other measurements. Particularly in the case of AI-related exposure scores, we anticipate that a combination of other measurements would have a strong correlation with our scores. However, earlier efforts had limited information about the future progress of LLMs or LLM-powered software. We expect that our understanding of future machine learning technologies is similarly imperfectly captured by our rubric today.